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Antibiotic tolerance: targeting bacterial survival

Lucas Boeck[1,2]

Antimicrobial susceptibility testing is the cornerstone of antibiotic treatments. Yet, active drugs are frequently unsuccessful in vivo and most clinical trials investigating antibiotics fail. So far, bacterial survival strategies, other than drug resistance, have been largely ignored. As such, drug tolerance and persisters, allowing bacterial populations to survive during antibiotic treatments, could fill a gap in antibiotic susceptibility testing. Therefore, it remains critical to establish robust and scalable bacterial viability measures and to define the clinical relevance of bacterial survivors across various bacterial infections. If successful, these tools could improve drug design and development to prevent tolerance formation or target bacterial survivors, to ultimately reduce treatment failures and curb resistance evolution.

Addresses
[1] Department of Biomedicine, University Basel, Basel, Switzerland
[2] Clinic of Pulmonary Medicine, University Hospital Basel, Basel, Switzerland

Corresponding author: Boeck, Lucas (lucas.boeck@unibas.ch)

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Introduction

Antibiotic susceptibility measures (e.g. the minimal inhibitory concentration (MIC)), testing or predicting antibiotic growth inhibition, are considered to define antibacterial activity. However, the performance of these tests, variable between infections, is far from perfect, and commonly treatment failures remain unexplained. For example, despite using one or more active drugs, the treatment of *Staphylococcus aureus* bacteraemia fails in ~40% of patients [1], ~10% of urinary tract infections recur within a month of treatment [2] and non-tuberculous mycobacterial infections can relapse in over 50% [3]. Similarly, insufficient tools to streamline drug candidates for clinical studies result in many trial failures, consuming hundreds of millions of USD and years of development [4].

Drug tolerance is a bacterial characteristic and strategy to succeed in hostile environments, which has been largely ignored, but could bridge the gap between in vitro assessments and clinical outcomes. Whereas drug resistance allows bacteria to grow during antibiotic exposure (increased MIC), drug tolerance is the ability of bacteria to survive transiently under normally lethal antibiotic conditions (unchanged MIC). Drug tolerance reflects a slower antibiotic kill rate of the main bacterial population (first slope of biphasic bacterial killing), whereas persisters are considered a highly tolerant subpopulation (second slope of biphasic killing). Both drug tolerance and persisters generate survivors that have the potential to re-establish the infection, that is, to cause relapses and treatment failures, once the drug is removed. In this opinion article, I summarise the current knowledge on antibiotic tolerance and address bottlenecks and opportunities for drug discovery.

Complex effects of antibiotics

The bacterial response to antibiotics or antimicrobial action is neither binary (susceptible vs. resistant) nor continuous (MICs) but more complex. We have long time assumed that bacterial populations are homogeneous, with rapid growth at low antibiotic concentrations that successively transition into growth inhibition and potentially killing at higher concentrations. However, bacterial populations are composed of individual bacteria that can behave very differently. Some bacteria may replicate at drug concentrations much higher than the overall MIC (heteroresistant populations [5]), whereas the persister subpopulation (also referred to as heterotolerant) survives drug concentrations killing most bacteria. Bacterial growth, growth inhibition and even killing might coexist at identical antibiotic drug concentrations, resulting in a dynamic state of bacterial growth and death [6]. Conventional drug susceptibility testing captures mainly bulk population growth and ignores more subtle antibiotic effects and bacterial responses. I propose a single-cell model of antibacterial activity (Figure 1, *animation*), describing bacterial growth and killing of different populations across many drug concentrations. Based on bacterial genealogy, this model illustrates antibacterial activity on a single-cell level, highlighting the complexity of drug effects in a single environment. At present, we can only study parts of this

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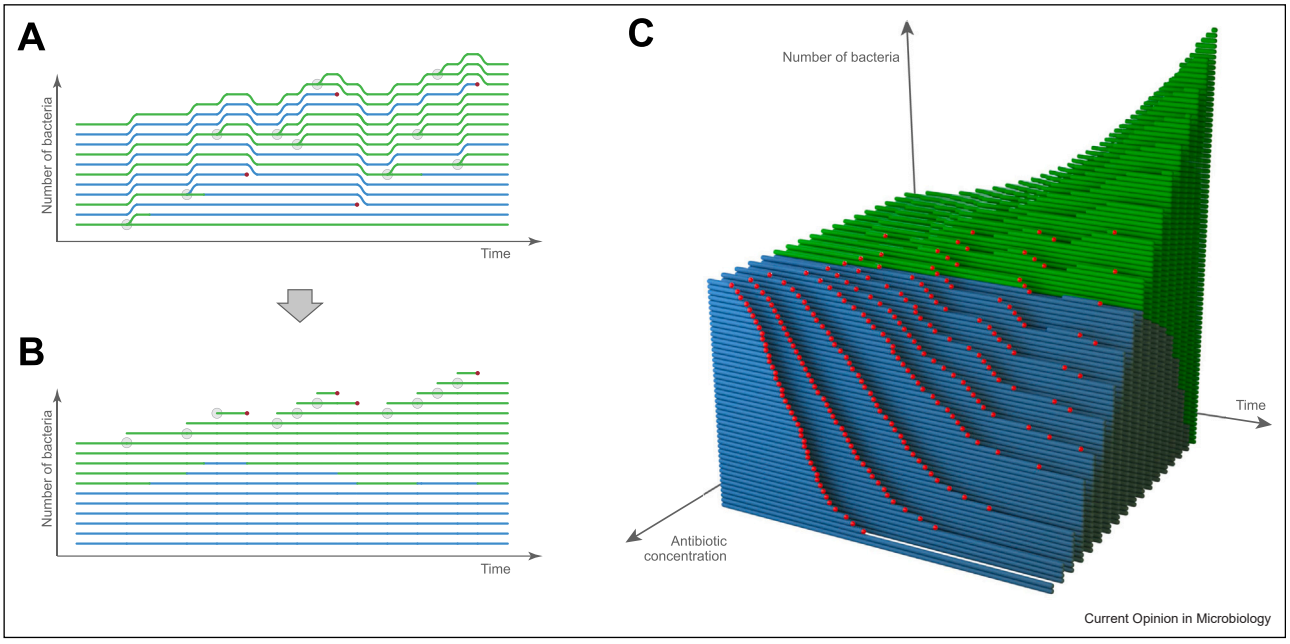
Figure 1



The 3D landscape of antibiotic action, integrating growth and killing across time and antibiotic concentrations. **(a)** Bacterial genealogy of single bacteria with growing (green; grey circle: bacterial division), non-growing (blue) and killed (red dot) bacteria during antibiotic treatment. **(b)** Simplified representation of single-cell growth and viability states (killed bacteria, growing and growth-arrested bacteria are reordered from top to bottom). **(c)** 3D illustration to highlight antibiotic action and bacterial responses across different drug concentrations on a single-cell level. This model captures drug resistance (growth at high antibiotic concentrations), heteroresistance (simultaneous growth and growth inhibition at multiple concentrations), drug tolerance (slow antibiotic killing of the total population) and persisters (a highly tolerant subpopulation).

model experimentally; however, technological advances (e.g. in live-cell imaging) will allow us to assess thousands of bacterial trajectories and population behaviours [7]. Some of these phenotypic traits are likely controlled by the bacterial genetic background and are therefore approachable with molecular diagnostic tools.

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Drug tolerance and persisters

Antibiotic killing, as suggested with bactericidal drugs, is not only defined by drugs but also driven by the bacterium in the context of a particular environment. Once antibiotics inhibit bacterial growth, the rate of bacterial killing is unrelated to drug resistance (MICs) and is thus a different characteristic of antimicrobial activity [8]. Presumably, bactericidal antibiotics have to corrupt ('poison' rather than inhibit) active targets in order to kill. Thus, target inactivity due to metabolic inactivity is linked with survival and is considered a state of impaired antibiotic lethality. Consequently, any bacterial state associated with slow growth or metabolic inactivity could facilitate drug tolerance or persister formation. These cell states are often characterised by energy depletion (adenosine triphosphate (ATP), proton motive force etc.) or activation of stress pathways. However, most dormant cells do not survive bactericidal drug concentrations [9,10], while some persisters remain metabolically active [11,12]. Therefore, tolerance and persisters can be generated by, but are not limited to metabolic inactivation [13].

The physiological changes underlying drug tolerance are driven by the environment, including the host and microbial communities, and the bacterial genetic background (Figure 2) [14,15]. While mechanisms of drug tolerance increase bacterial survival of the total bacterial population, persister mechanisms are different, as they generate bacterial heterogeneity and promote highly drug-tolerant subpopulations [13]. But despite the differences of tolerance and persisters, there also exist considerable overlaps in their mechanisms and behaviours. Drug tolerance and the persister fraction, similar to resistance, is genetically modifiable, highlighted with high-tolerance and high-persister mutants (high-tolerance mutants: slow killing of the bulk population; high-persister mutants: large subpopulation of highly tolerant cells). The persister state is generally considered non-heritable, with persisters forming a heterogeneous, mainly susceptible population after regrowth. Whether epigenetic modifications could preserve a 'persister memory' is unclear [16,17]. But importantly, the killing rate (drug tolerance) and the fraction of persisters

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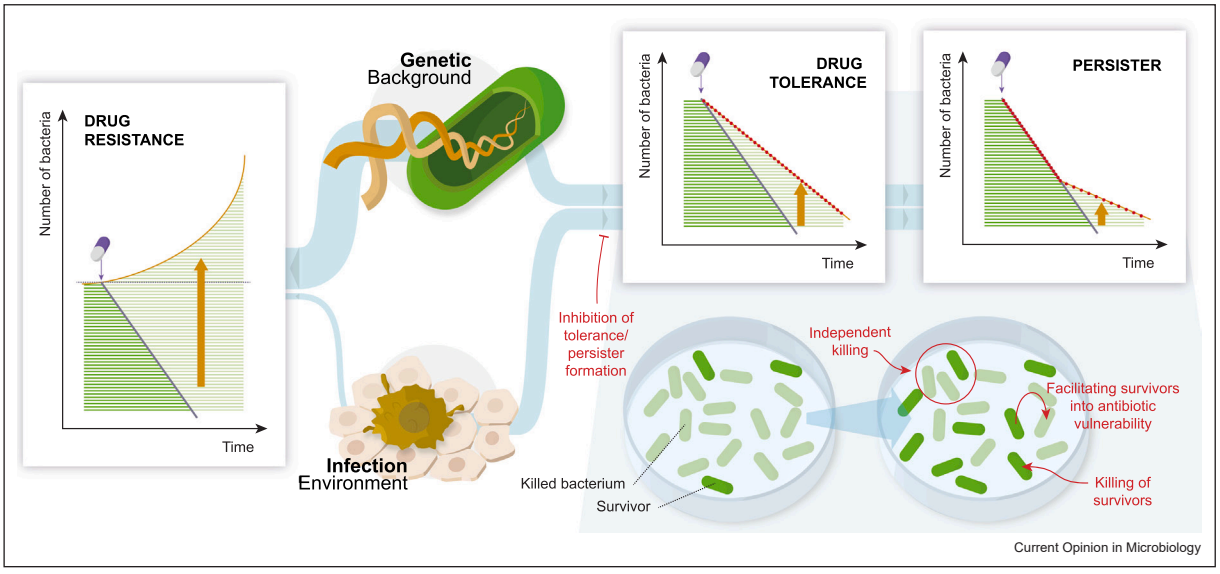
Figure 2



Causes and consequences of drug tolerance and persisters and potential therapeutic interventions. Drug resistance (the ability to grow in the presence of antibiotics) is mainly driven by the genetic makeup of bacteria. Drug tolerance (lower killing rate of the total population) and persister (highly tolerant subpopulation) formation are shaped by the genetic background and the environment. Drug tolerance and persisters increase the number of bacterial survivors. While drug tolerance is quantified by the initial killing slope, the second slope (biphasic killing) indicates persisters. These survivors could be addressed by blocking survivor formation, targeting and clearing survivors or facilitating them into vulnerable states. Targeting drug tolerance or persisters may also impair the evolution of drug resistance.

are heritable and selectable traits, where selective pressures and fitness trade-offs (tolerance can affect the bacterial growth rate, virulence and transmissibility) during host infection and therapy shape the evolution of drug tolerance and persisters [18–21]. As such, drug tolerance and persisters increase bacterial survival in usually lethal antibiotic conditions, in which survivors remain with the full infectious potential. These survivors create a reservoir for relapses and promote via an increased population size and potentially antibiotic-induced DNA damage, drug resistance and virulence [21–24].

Relevance of drug tolerance and persisters

Evidence that poor killing is the cause of treatment failures and infection relapses is limited. It remains challenging to unravel whether bacterial, host, pharmacokinetic or other factors result in treatment failures. However, several indirect cues exist that drug tolerance drives poor treatment outcomes. Most recurrent infections are drug-susceptible and genetically highly similar to the initial infection, indicating relapses rather than reinfections [25]. The anatomic sites of relapses are commonly the sites of known bacterial survival, such as mesenteric lymph nodes in *Salmonella* or pulmonary cavities in mycobacterial infections [26]. More evidence comes from studies showing that drug-tolerant bacteria and high-persister mutants are enriched and selected during infection and antibiotic treatments [20,27]. Smaller studies have also suggested an association between drug tolerance and in vivo outcomes [28–30]. Notably, the relevance of drug tolerance likely differs across infection environments and bacterial species. But given that we lack routine drug tolerance testing and efficient methods, the contribution of drug tolerance to treatment failures across various infections remains unknown and is probably the most significant gap in the antibiotic tolerance field. Designing, developing and testing tolerance-tailored treatments requires extensive efforts and unlikely translates into improved patient outcomes in infections where drug tolerance is irrelevant, but provides tremendous opportunities for conditions in which drug tolerance drives treatment failures.

Measures of drug tolerance

An alternative solution to proving that tolerant bacteria cause poor clinical outcomes, which is due to the small numbers of surviving bacteria inherently difficult, is to assess if in vitro tolerance measures account for unexplained treatment failures. But given that different, sometimes poorly standardised bacterial viability

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assessments exist, the chosen method will affect the performance for predicting clinical outcomes [31]. Consequently, a strategy linking in vitro measures to clinical outcomes would also benchmark the used tests for drug development and treatment approaches. Drug tolerance measures rely on viability definitions and assessments. Generally, bacteria that have the capacity to replicate are considered alive, whereas bacteria that are damaged beyond a 'point of no return' should be regarded as dead. While this distinction is not perfectly accurate (bacteria may contribute to a society without being able to replicate), it is even more challenging, if not impossible, to experimentally assess these viability states. There exists no perfect bacterial viability test, but rather surrogates of live and dead bacteria. Usually, bacterial viability is assessed with colony-forming units (CFUs), which are the gold standard of bacterial viability assessments and can be considered a 'live marker' where bacteria that form colonies are obviously alive. However, heterogeneous lag times, dormancy, post-antibiotic effects, specific media requirements or even toxicity due to the transfer of stressed bacteria to nutrient-rich media may interfere with our assumption that bacteria that do not form colonies are dead [32]. Similarly, metabolic activity (assessed with luminescence, ATP etc.) indicates that bacteria are alive, but viability remains ambiguous in case of metabolic inactivity. In contrast, other markers (e.g. dyes that stain cells with a disrupted membrane) can be considered 'dead markers' that mostly indicate unrepairable cell damage, with less knowledge of the functionality of the remaining cells. Another limitation is the poor scalability and cost of CFUs, restricting the study and applicability of drug tolerance measures. Therefore, several new approaches emerged that miniaturise, automate and modify existing tests or propose new solutions to assess bacterial killing at a large scale [33–36]. For example, the tolerance disk test (TDtest) replaces the disk of the disk-diffusion test after antibiotic treatment with a nutrient disk to reveal bacterial regrowth and survival. Once clinically validated, such methods can become critical to advance killing measures into drug design, development and tolerance-tailored treatments.

Designing anti-tolerance/persister treatments

Bacterial survival is linked to cellular processes that render antibiotics less lethal. Preventing tolerance and persister formation is considered difficult since these cellular states are formed very early in infection due to stochastic persister generation, nutrient deprivation, hostile infection environments, immunity or other stimuli [37]. However, drug tolerance and persisters can also emerge during antibiotic treatment. Therefore, protective mechanisms, such as the SOS response or stringent response enzymes, could be targeted to disrupt antibiotic-induced tolerance and persister formation [38]. Moreover, persisters and populations of highly tolerant bacteria also arise during chronic infections (e.g. infections in cystic fibrosis patients), where more stable conditions alternate with acute infection flares. Interfering with tolerance formation could continuously decrease the survivor reservoir and promote infection clearance. A major obstacle is the redundancy and diversity of tolerance mechanisms, creating multifaceted survivors rather than a homogeneous survivor population [39]. Optimally, we would identify and target chokepoints critical for multiple tolerance pathways. For example, hydrogen sulphide protects bacteria from oxidative stress, whereas pharmacologic disruption of bacterial hydrogen sulphide synthesis prevented tolerance formation and potentiated bacterial killing in *Staphylococcus aureus* and *Pseudomonas aeruginosa* [40]. Targeting tolerance pathways may prevent tolerance formation and kill survivors once formed.

Another strategy is to kill bacteria independent of their cellular processes. Considering the bacterial metabolic effects on antibiotic lethality, it was proposed to assess if bactericidal action is preserved across different metabolic conditions [41,42]. While antibiotics with strong metabolic dependencies kill mainly metabolically active bacteria, antibiotics with weak metabolic dependencies have a relatively energy-independent lethal action. These drugs target mechanisms and structures, such as the DNA or cell membrane, that are also required for dormant cells to remain viable [43]. Mitomycin C or cisplatin, for example, cross-links DNA and kills persisters of multiple species. Antimicrobial peptides or lysins, such as the diaryl urea derivate PQ401, have profound membrane-damaging activity and kill drug-tolerant bacteria [44,45]. Similarly essential is the transmembrane proton motive force and energy homoeostasis of bacteria, targeted with compounds like bedaquiline and econazole [46,47].

A different approach is to hijack bacterial functions to kill bacteria. Hydrolases are particularly suitable since hydrolysis does not require energy and can degrade essential molecules. Compounds with an anti-persister activity that target hydrolases are lassomycin, which causes energy-independent ATP hydrolysis and depletion, and acyldepsipeptide (ADEP), which activates protein hydrolysis and damages proteins. Notably, to corrupt intracellular targets, drugs need to penetrate the bacterial cell wall. During metabolic inactivity and reduction in proton motive force, active uptake (e.g. uptake of aminoglycosides) is arrested, and diffusion through porins is reduced [48]. Conversely, drugs that passively diffuse into cells and are typically removed with active export systems accumulate within metabolically silent bacteria. Such drugs (e.g. minocycline) can,

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due to their specific pharmacokinetics, target dormant bacteria [49]. Survivors can also be sensitised to become vulnerable to conventional antibiotics due to activating cells and their targets or antibiotic uptake. Several compounds such as adenosine and thymine facilitate metabolic activity and consequently promote bacterial killing [50,51], whereas other compounds increase proton motive force and antibiotic influx in dormant bacteria.

Overall, these approaches have the potential to reduce or clear a very small fraction of bacteria that survive and are likely responsible for treatment failures and relapses, as well as long and sometimes toxic treatments. Therefore, targeting drug tolerance and persisters could improve clinical outcomes and shorten antibiotic treatment. Moreover, by reducing the bacterial population size and limiting opportunities for DNA damage, such therapies may mitigate the emergence of drug resistance and thereby increase the lifetime of available drugs.

Conclusion

Our strategies for developing and applying antibiotic treatments are hampered by the ignorance of bacterial strategies to overcome toxic stresses. Drug tolerance and persisters increase bacterial survivors, which have the potential to re-establish the infection after treatment. Characterising drug tolerance may reduce the gap between conventional antibiotic in vitro measures and clinical outcomes and could allow to optimise treatments and reduce clinical trial failures. Moreover, treatments targeting drug tolerance and persisters, or bacterial survivors in general, could lead to highly efficient and much shorter antibiotic treatments.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationship that could have appeared to influence the work reported in this paper.

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- •• of outstanding interest

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